

# **Reducing Cardiovascular Disease in Indonesia**

## **Health Impact, Costs, and Cost-Effectiveness of Clinical and Public-Health Interventions, 2025–2050**

World Bank · University of Washington

Briefing for the Ministry of Health, Republic of Indonesia

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## **Reducing CVD in Indonesia**

# Why this matters for Indonesia

Cardiovascular disease is Indonesia's largest health burden

Under current trends, a person reaching age 30 faces roughly a **14.5% chance** of dying from cardiovascular disease before age 70 — the CVD “40q30”. Most of this is **preventable or treatable**.

- Premature CVD deaths (ages 30–69) have risen for two decades and remain high.
- Two complementary levers:
  - **Clinical care** — prevent and better treat established disease.
  - **Population prevention** — tobacco, salt, trans-fat.
- This analysis weighs their **health impact and cost-effectiveness** together.

Source: Indonesia CVD model (calibrated to GBD 2023; UN World Population Prospects 2024). CVD 40q30 = probability of dying from cardiovascular disease between exact ages 30 and 70.

# What this analysis set out to do

## General aim

Estimate the **health impact and cost-effectiveness** of scaling up evidence-based **clinical** and **public-health** cardiovascular interventions in Indonesia through **2050**.

### Aim 1 — Burden

Project the CVD burden and the premature-mortality indicator (CVD 40q30) under the **status quo** and under **alternative intervention scenarios**.

### Aim 2 — Value for money

Estimate **deaths averted**, mortality reductions, **incremental costs**, and the **cost per death averted** for each intervention and package.

### Aim 3 — Better together

Quantify the **added value of combining** clinical and public-health action, modeled as a single joint run rather than a sum.

Horizon 2025–2050; the “status quo” baseline is the no-new-intervention comparator against which every scenario is measured.

# Executive summary — the headline numbers

## Combining clinical + public health is the big prize

Together they could avert about **3.5M deaths** over 2025–2050, cut the CVD 40q30 in 2050 by **29%**, at about **US\$18,486 per death averted**.

- **All clinical interventions:**  $\approx$ 2.1M deaths averted; CVD 40q30 down 20% by 2050.
- **All public-health policies:**  $\approx$ 1.6M deaths averted at about US\$305 per death averted.
- **Most impactful single levers:** **Primary CVD prevention** and **Tobacco tax**.
- **Affordability:** the combined package costs about US\$18,486 per death averted.

All figures from the model's current-run cost/value outputs. Deaths averted are cumulative 2025–2050 across modeled cardiovascular and diabetes causes. Cost per death averted uses discounted incremental cost in market US\$ (price year 2023).

# How the model works — in plain language

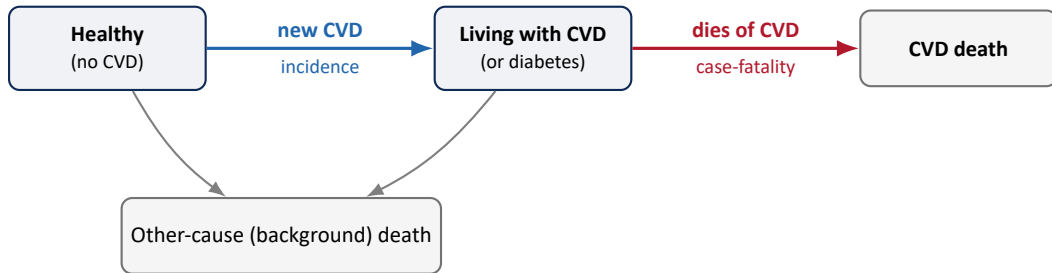
1. **People & demography.** Indonesia's population is projected by age and sex (UN World Population Prospects 2024).
2. **Starting health.** Disease and death rates are set to Indonesia's data (GBD 2023) and calibrated to match observed levels.
3. **Year-by-year simulation.** Each year people can develop cardiovascular disease or diabetes, survive, or die.
4. **Interventions.** Clinical and public-health scenarios change how many people develop disease and how many survive it.
5. **Outcomes.** The model reports deaths, healthy years (DALYs), premature-mortality (40q30), costs, and cost-effectiveness.

## One consistent engine

Every scenario runs through the **same** projection, so results are directly comparable and the **combined** package is modeled as one run — not a sum of separate estimates.

Calibration fits the model to GBD estimates using a standard numerical optimiser. Demography and life expectancy use UN WPP 2024.

# How interventions change the disease pathway



## Fewer people fall ill

Primary prevention & *all* public-health policies lower **incidence**.

## More people survive

Secondary prevention, treatment & tobacco policy lower **case-fatality**.

## Background mortality

Other-cause deaths from demography; unchanged by these interventions.

No recovery/remission transition is modeled: people living with CVD remain at elevated risk (a conservative assumption).

Source: model engine, 06\_run\_scenarios\_indonesia\_fair.R.

## **The interventions**



# Clinical interventions (FAIR Choices)

| Intervention                      | Target population                | Model effect  |
|-----------------------------------|----------------------------------|---------------|
| Primary CVD prevention            | Adults at cardiovascular risk    | Incidence     |
| Type-2 diabetes treatment         | People with type-2 diabetes      | Case-fatality |
| Heart-attack secondary prevention | Heart-attack (IHD) survivors     | Case-fatality |
| Chronic heart-failure care        | Chronic heart-failure patients   | Case-fatality |
| Acute heart-failure care          | Acute heart-failure patients     | Case-fatality |
| Stroke secondary prevention       | Stroke survivors                 | Case-fatality |
| RHD secondary prevention          | Rheumatic heart disease patients | Case-fatality |

**Implementation scenario (all clinical interventions):** coverage scales up **from a low base toward 80%** of the eligible population between 2025 and 2050. Effect sizes are drawn from the FAIR Choices intervention taxonomy; primary prevention lowers who develops CVD, while secondary prevention and treatment improve survival. Source: workbook Selected\_Interventions.

# Public-health interventions (fiscal & regulatory policy)

| Policy                 | Risk factor         | Model effect         | Implementation lever                         |
|------------------------|---------------------|----------------------|----------------------------------------------|
| Tobacco tax            | Tobacco             | Incidence + survival | Raise tax to 75% of retail price             |
| Sodium policy package  | Dietary salt/sodium | Incidence            | Reformulation, labelling, media, procurement |
| Tobacco policy package | Tobacco             | Incidence + survival | Clean-air law, mass media, ad ban            |

Policies reduce exposure (smoking, salt, trans-fat intake), lowering CVD **incidence**. **Tobacco** policies additionally improve **survival** after CVD. Packages are modeled as single joint runs of their component measures. Source: workbook `Selected_Interventions / Policy_Lever`s.

# Data sources

| Model component                          | Primary source                               | Years / version | Use in model                                      |
|------------------------------------------|----------------------------------------------|-----------------|---------------------------------------------------|
| Population & demography                  | UN World Population Prospects                | 2024 revision   | Population, background mortality, life expectancy |
| Cause-specific mortality & disease rates | GBD (Indonesia)                              | 2023            | Baseline burden & calibration targets             |
| Disability weights (healthy-years lost)  | GBD                                          | 2023            | DALYs                                             |
| Clinical intervention effects            | FAIR Choices intervention taxonomy           | current         | Incidence & case-fatality effects                 |
| Public-health effect sizes               | Published risk-factor evidence (in workbook) | current         | Incidence & survival effects                      |
| Intervention & policy costs              | Programme & policy unit costs (in workbook)  | price year 2023 | Budget impact & cost-effectiveness                |
| International 40q30 comparison           | WPP-scaled ASEAN comparison series           | 1995–2050       | Benchmarking Indonesia vs. neighbour              |

Sources verified in the modeling pipeline and input workbooks. Specific effect-size and cost citations are documented in the input workbooks. Population: UN World Population Prospects 2024; GBD (GBD 2023 Cardiovascular Diseases and Risks Collaborators 2025; Schumacher et al. 2024).

## **Health impact and value for money**

## Clinical: impact and cost-effectiveness

| Clinical intervention             | Deaths averted<br>(cumulative) | CVD 40q30<br>2050 | 40q30 red.<br>2050 | Deaths red.<br>2050 | Cost per<br>death averted |
|-----------------------------------|--------------------------------|-------------------|--------------------|---------------------|---------------------------|
| Primary CVD prevention            | 1,322,000                      | 7.6%              | 16.5%              | 10.9%               | US\$919                   |
| Type-2 diabetes treatment         | 463,000                        | 9.0%              | 0.4%               | 3.0%                | US\$127,368               |
| Heart-attack secondary prevention | 141,000                        | 8.9%              | 1.8%               | 0.8%                | US\$21,376                |
| Chronic heart-failure care        | 77,000                         | 9.0%              | 0.9%               | 0.5%                | US\$7,831                 |
| Acute heart-failure care          | 69,000                         | 9.0%              | 0.6%               | 0.5%                | US\$13,210                |
| Stroke secondary prevention       | 31,000                         | 9.1%              | 0.3%               | 0.2%                | US\$56,216                |
| RHD secondary prevention          | 6,000                          | 9.1%              | 0.0%               | 0.0%                | US\$24,755                |
| <b>All clinical (combined)</b>    | <b>2,072,000</b>               | <b>7.3%</b>       | <b>19.9%</b>       | <b>15.5%</b>        | <b>US\$31,733</b>         |

Cost per death averted is based on **discounted** incremental costs (3% per year) and assumes a linear scale-up from baseline coverage in 2025 to 80% coverage by 2050. Costs are reported in market US\$ (price year 2023) from the health system perspective.

# Public health: impact and cost-effectiveness

| Public-health policy         | Deaths averted<br>(cumulative) | CVD 40q30<br>2050 | 40q30 red.<br>2050 | Deaths red.<br>2050 | Cost per<br>death averted |
|------------------------------|--------------------------------|-------------------|--------------------|---------------------|---------------------------|
| Tobacco tax                  | 724,000                        | 8.6%              | 5.2%               | 3.3%                | US\$283                   |
| Sodium policy package        | 540,000                        | 8.7%              | 4.0%               | 3.0%                | US\$333                   |
| Tobacco policy package       | 435,000                        | 8.8%              | 3.1%               | 2.0%                | US\$272                   |
| All public health (combined) | 1,649,000                      | 8.0%              | 11.9%              | 8.0%                | US\$305                   |

Cumulative deaths averted during 2025–2050, relative to the status-quo baseline. The tobacco package combines smoke-free policies, health warnings and mass-media campaigns, advertising bans, and cessation support. The sodium package combines food reformulation, front-of-pack nutrition labelling, healthy public food-procurement policies, and behavior-change communication. Each package is modeled as a single joint scenario.

## **The combined package**

## Clinical + public health: better together

| Package                                    | Deaths averted<br>(cumulative) | 40q30 red.<br>2050 | CVD 40q30<br>2050 | Total cost<br>(PV, discounted) | Cost per<br>death averted |
|--------------------------------------------|--------------------------------|--------------------|-------------------|--------------------------------|---------------------------|
| All clinical (combined)                    | 2,072,000                      | 19.9%              | 7.3%              | US\$65.8B                      | US\$31,733                |
| All public health (combined)               | 1,649,000                      | 11.9%              | 8.0%              | US\$503.2M                     | US\$305                   |
| <b>Clinical + public health (combined)</b> | <b>3,547,000</b>               | <b>29.2%</b>       | <b>6.4%</b>       | <b>US\$65.6B</b>               | <b>US\$18,486</b>         |

### The combined package is modeled as one run — not a sum

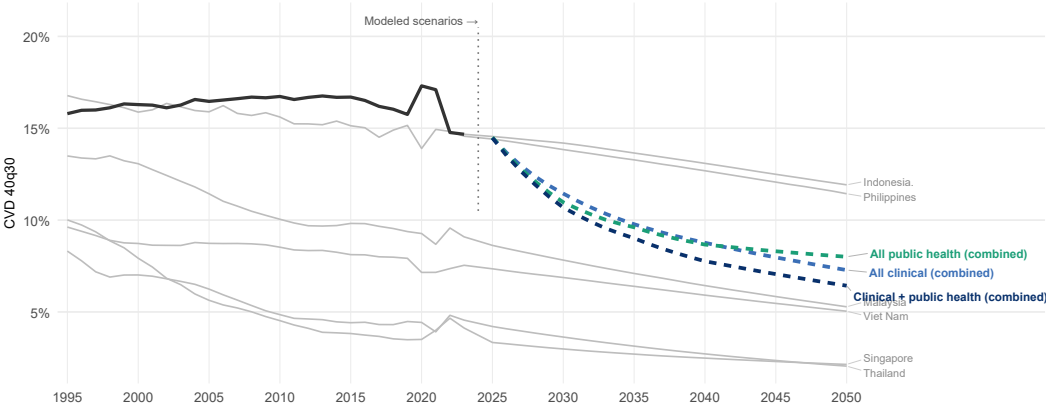
Because clinical and public-health effects overlap, the joint scenario is projected in a single run. It averts about **3.5M deaths**, brings the 2050 CVD 40q30 down to **6.4%** (from 9.1% at baseline), at about **US\$18,486 per death averted**.



# CVD premature mortality: trajectory and neighbours

## CVD 40q30 in Indonesia: observed, projected, and neighbours

Probability of dying from CVD between ages 30 and 70



Indonesia observed history (solid, to 2023) and ASEAN comparators are WPP-scaled reference series.  
Modeled intervention scenarios are dashed for 2025–2050.  
ASEAN comparator trends use linear extrapolation for illustrative reference.

## **What it means for Indonesia**

# Policy implications

- **Biggest health gains come from doing both.** The combined package averts far more (~3.5M) than either family alone.
- **Public-health policies are exceptional value.** Population-wide fiscal/regulatory measures avert deaths at a fraction of clinical cost per death averted.
- **Clinical scale-up delivers large, durable gains** in survival and premature mortality.
- **Sequence for early wins.** Tobacco taxation and other prevention policies can start cutting risk quickly and cheaply, while clinical capacity scales up.
- **The 40q30 target is within reach.** The combined package bends Indonesia's premature-CVD-mortality curve down by about 29% by 2050.

These are model findings; translating them into policy requires judgement on feasibility, financing and delivery. Model results quantify the *potential*; they are not a delivery guarantee.

# Limitations

- Results depend on **modeled epidemiological estimates** (GBD) and calibration; real burden may differ.
- **Effect sizes, future coverage, and unit costs are uncertain**; interventions are assumed to reach and sustain their targets.
- The disease model is **simplified** (healthy → living-with-CVD → death; no remission; no comorbidity interactions).
- Cost-effectiveness is expressed as **cost per death averted**; morbidity and productivity effects are not monetised here.
- Projections are **scenarios, not forecasts or guarantees**; they show what is achievable if interventions are delivered as modeled.

# Three things to remember

## 1. Both, together

Clinical + public health averts **~3.5M deaths** by 2050 — more than either alone.

## 2. Prevention is affordable

Population policies avert deaths at about **US\$305** each.

## 3. A reachable target

Premature CVD mortality (40q30) falls **29%** by 2050 under the combined package.

Combined action against CVD could avert about **3.5M deaths** at about **US\$18,486 per death averted** through 2050.

## Technical appendix

# Appendix 1 — Scenario crosswalk

| Family                         | Workbook scenario id       | Executive label                                       |
|--------------------------------|----------------------------|-------------------------------------------------------|
| <b>Clinical (FAIR Choices)</b> | all                        | All selected interventions (combined)                 |
|                                | I_CVD_PRIMARY              | Primary prevention of CVD                             |
|                                | I_HF_ACUTE                 | Acute heart-failure management                        |
|                                | I_HF_CHRONIC               | Chronic heart-failure management                      |
|                                | I_IHD_SECONDARY            | Secondary prevention of ischemic heart disease        |
|                                | I_RHD_SECONDARY            | RHD secondary prevention                              |
|                                | I_STROKE_SECONDARY         | Secondary prevention of stroke                        |
|                                | I_T2D_TREATMENT            | Longitudinal treatment of type 2 diabetes             |
|                                | baseline                   | Baseline (no new intervention)                        |
|                                |                            |                                                       |
| <b>Public health</b>           | all_public_health          | All selected public-health interventions (combined)   |
|                                | I_PH_SALT_POLICY           | Salt policy package                                   |
|                                | I_PH_TOBACCO_POLICY        | Tobacco policy package                                |
|                                | I_PH_SALT_ENV              | Supportive public-institution environment             |
|                                | I_PH_SALT_FOPL             | Sodium front-of-pack labelling                        |
|                                | I_PH_SALT_MEDIA            | Sodium mass-media campaign                            |
|                                | I_PH_SALT_REFORM           | Mandatory sodium reformulation                        |
|                                | I_PH_TOBACCO_TAX           | Tobacco tax                                           |
|                                | I_PH_TOB_AD_BAN            | Tobacco advertising ban                               |
|                                | I_PH_TOB_CLEAN_AIR         | Smoke-free / clean-air law                            |
|                                | I_PH_TOB_MEDIA             | Tobacco mass-media campaign                           |
|                                |                            |                                                       |
| <b>Combined</b>                | all_clinical_public_health | All clinical + public-health interventions (combined) |

# Appendix 2 — Definitions, horizons and assumptions

## Metrics

- **Deaths averted (cumulative)**  
$$= \sum_{t=2025}^{2050} (\text{baseline deaths}_t - \text{scenario deaths}_t),$$
  
across modeled CVD & diabetes causes.
- **% reduction in deaths, 2050**  
$$= 100 \times (\text{baseline} - \text{scenario}) / \text{baseline in 2050}.$$
- **CVD 40q30** = probability of dying of CVD between exact ages 30 and 70 (six CVD causes, both sexes, single-year life table).
- **% reduction in 40q30, 2050**: analogous to deaths.

**Model.** Discrete-time state-transition model; single-year ages 0–95; six CVD causes (IHD, ischemic stroke, intracerebral haemorrhage, hypertensive heart disease, rheumatic heart disease, cardiomyopathy & myocarditis) plus type-2 diabetes; calibrated to GBD 2023; demography and life expectancy from UN WPP 2024. **Data provenance:** every number is read from the current-run results contract (output/09\_deck\_results.rds) and the workbooks' literal CVD 40q30 series; scenario identity follows the Scenario\_Catalog contract (binding include\_flag). No Excel recalculation is used.

## Costing & cost-effectiveness

- Horizon 2025–2050; cost discount rate 3%/yr; price year 2023.
- Costs in **market US\$**; cost perspective: health system.
- **Cost per death averted** = cumulative **discounted** incremental cost ÷ cumulative **undiscounted** deaths averted.
- A “Dominant” scenario averts deaths while lowering cost; “not defined” means no deaths averted.



## Appendix 3 — References

GBD 2023 Cardiovascular Diseases and Risks Collaborators. 2025. “Global, Regional, and National Burden of Cardiovascular Diseases and Risk Factors in 204 Countries and Territories, 1990–2023.” *Journal of the American College of Cardiology* 86 (22): 2167–2243.

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